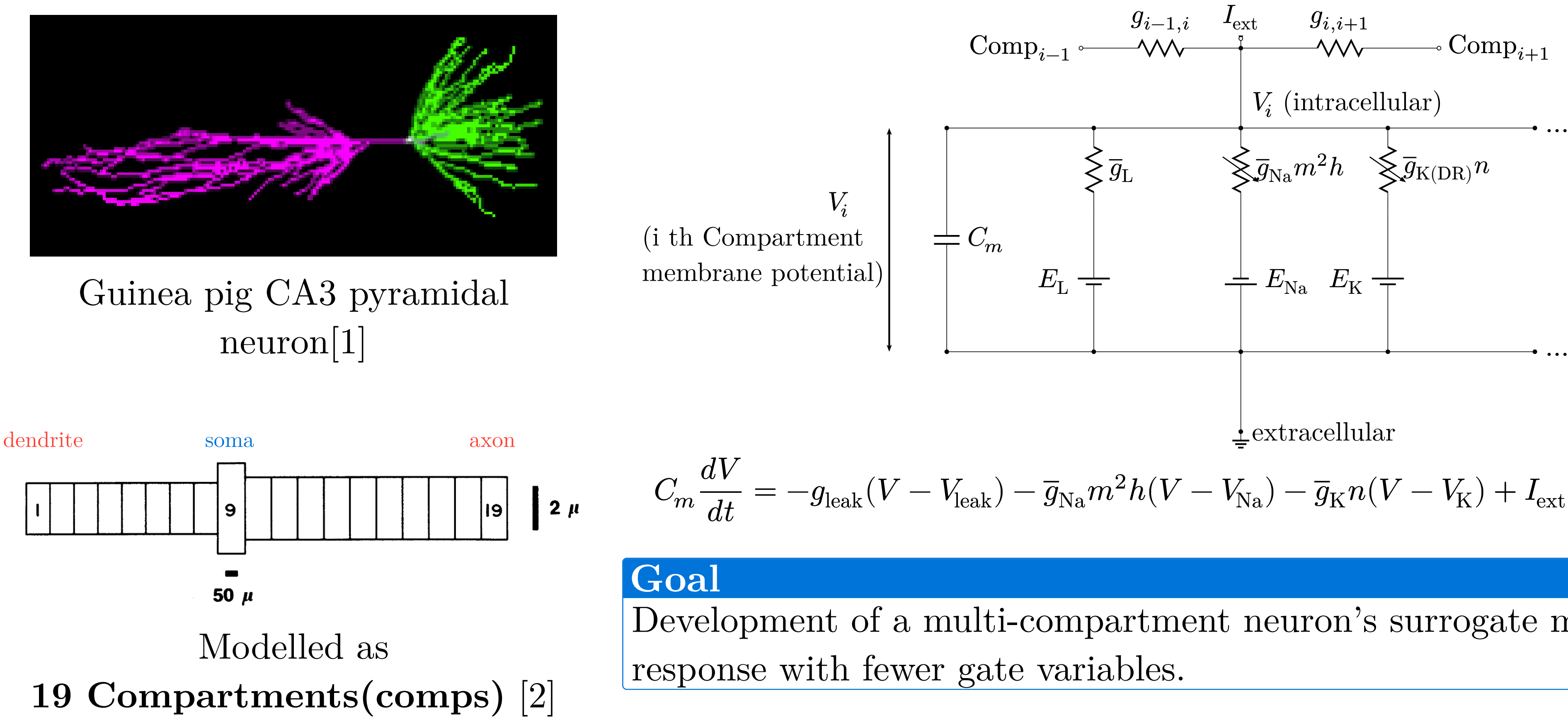


Introduction



Conductances have **gate variables**, ODEs with rate functions α, β :

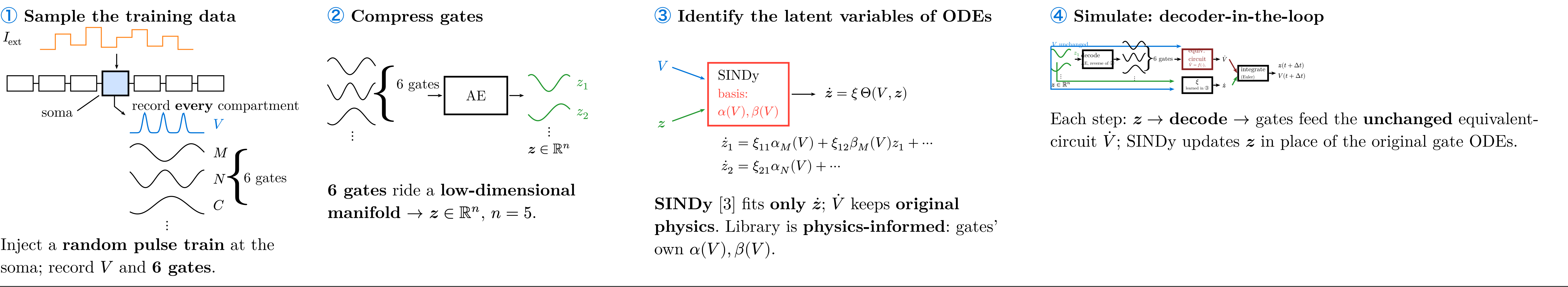
$$I_{\text{Na}} = \bar{g}_{\text{Na}} m^2 h (V - E_{\text{Na}})$$
$$\frac{dm}{dt} = \alpha_m(V)(1 - m) - \beta_m(V)m$$
$$\alpha_m(V) = 0.32(13.1 - u)/(\exp((13.1 - u)/4) - 1)$$
$$\beta_m(V) = 0.28(u - 40.1)/(\exp((u - 40.1)/5) - 1)$$

→ **11 states per compartment**
→ **209** for 19 comps;
At large scale network simulation, the gates become a **memory bottleneck**.

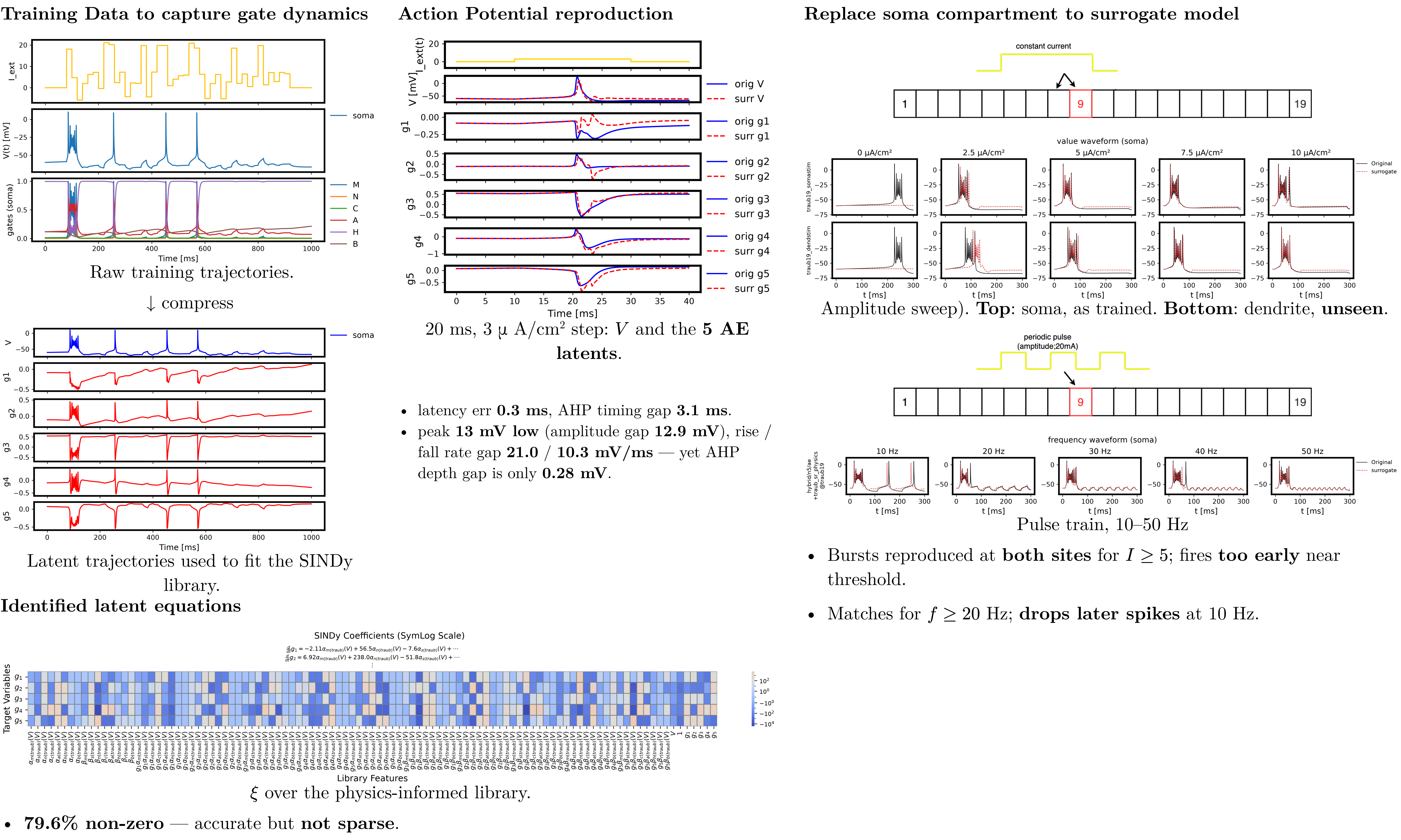
Goal

Development of a multi-compartment neuron’s surrogate model capable of reproducing the membrane potential response with fewer gate variables.

Methods



Results and Discussion



Conclusion

- Latent dynamics preserve **timing**.
- Error concentrates in the **fast, steep transition** — suggests the **AE encoding (n=5)**, not the SINDy library, fails to preserve it.
- Replaced **all 19 compartments**, no divergence; transfers to **unseen site • frequency**.
- **Core finding**: SINDy succeeds **not because we compressed (n: 6→5)** — ⑤ shows the **same result uncompressed (n = 6)**, so it is the **coordinate change itself** that makes gate dynamics learnable.
- Cost reduction follows **only if** compression matters — so it remains **unproven**: ξ is still dense (**79.6%** non-zero).

Code — github.com/MunechikaHaruki/SINDyNeuroSurrogate

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→ **Future:** sparsify ξ , push n down, test on other channel models.

Conflicts of Interest — The authors declare no conflicts of interest regarding this manuscript.